

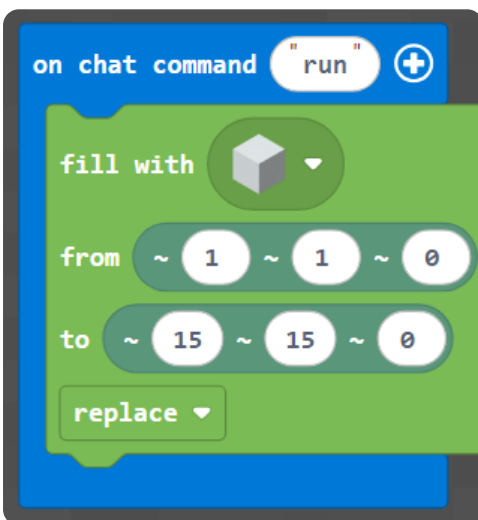
M003 Week 4 — English Worksheet (Beginner)

Topic: Pixel Art 2 — Cat & Tree (x, y) · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

This week you copy **two more pictures** — a **cat face** and a **tree**. Every block sits at a spot named by **two numbers** — (x, y). x is how far **across**, y is how far **up**. You read the picture and place each block at its (x, y).

First, build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
```

● **Stand on the red block every time you run.** If you move, the picture moves too and the blocks land in the wrong place. Same spot every time = a clean picture.

1 · Predict 🧠

Read the steps. Before you imagine it happening, circle or write your answer.

```
place brown block at (8, 1)
place brown block at (8, 2)
place brown block at (8, 3)
```

Do these blocks make a line going up? Circle one: across · up

Which number stays the same — x or y ? (This is the tree trunk.)

place orange block at $(2, 15)$

place orange block at $(3, 15)$

These share $y = 15$ and x grows. Do they go across or up? Circle one: across · up

2 · Find the Difference 🐛

Each pair shows the **correct** line first, then a **broken** one. Circle what's different and write one short sentence about the bug.

Pair A — The tree trunk should grow **up** at $x = 8$: (8, 1), (8, 2), (8, 3).

```
# correct
place brown block at (8, 1)
place brown block at (8, 2)
place brown block at (8, 3)
```

```
# broken
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
```

What is wrong? Where do all three blocks land?

Pair B — The trunk should grow **up** at $x = 8$.

```
# correct
place brown block at (8, 2)
place brown block at (8, 3)
```

```
# broken
place brown block at (2, 8)
place brown block at (3, 8)
```

What is wrong? Which two numbers got swapped?

3 · Fill the Gap

To grow the trunk **up**, the **second** number (y) has to grow. Fill in the missing numbers so the trunk goes up.

```
place brown block at (8, 1)
place brown block at (8, __)
place brown block at (8, __)
```

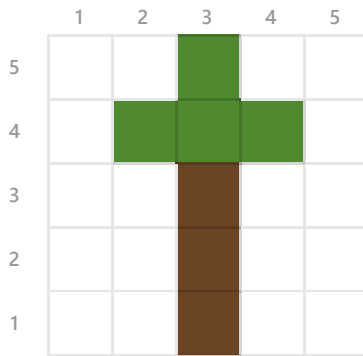
Word bank for the blanks: 2 · 3

Write the two missing numbers:

home spot (the red block).

Warm up first. Copy this small tree onto your wall. Read each square's (x, y) and place that color there.

 Small tree



Now pick one big picture — the 🐱 cat face or the 🌳 tree from the main worksheet grid — and copy it the same way. Read each colored square and place that color at its (x, y).

When you finish, send a photo of your picture, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

I built the ...

The lowest block in my picture is at $(x = \dots, y = \dots)$.

One tricky moment was when ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while you show your picture on the wall. Talk like you are teaching a friend. Try to use these words: **x, y, across, up, home spot**.

1. Show your home spot (the red block) and say why you stand on it.
2. Show your finished picture.
3. Point at one block and read its coordinate: "this one is at x ..., y ...".

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.